MIL-W-45223A(MR)
28 June 1968

SUPERSEDING
MIL-W-45223(MR)
26 June 1962

MILITARY SPECIFICATION

WELDING, SPOT, HARDENABLE STEELS

1. SCOPE

1.1 Scope. This specification covers requirements for spot welding carbon or low alloy steels. Steels of this type having greater than 0.20 percent carbon content are considered within the scope of this specification.

1.2 Classification. Classification shall be in accordance with class A, B, or C of MIL-W-6858 as specified in the contract or order or on the applicable drawing.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-W-6858 - Welding, Resistance, Aluminum, Magnesium, Non-hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. EQUIPMENT AND PROCESS REQUIREMENTS

3.1 The equipment and processes employed to accomplish spot welding as required by this specification shall be subject to the approval of the Government as specified in MIL-W-6858.

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3.1.1 Equipment. Equipment shall be in accordance with the requirements specified in MIL-W-6858 and the additional requirements that provisions shall be made for tempering the spot welds either in the welding machine or in a furnace after welding.

3.1.2 Tests of equipment. Qualification tests of the welding equipment and certification tests of the welding schedules shall be in accordance with MIL-W-6858 except for the addition of tensile tests using the tensile pullout specimens shown in figure 1. The jig to be used for testing the tensile pullout specimens shall be in accordance with figure 2. When the spot welds are tempered by a heating sequence in the welding machine, 10 tensile pullout specimens for each gage combination shall be made and tested in addition to the shear specimen requirements in MIL-W-6858. Tensile pullout specimens are not required when the spot welds are tempered or heat treated in a furnace after welding provided the tempering temperature is equal to or above that normally used in the heat treatment of the parent metal.

4. SPOT WELD REQUIREMENTS

4.1 Shear specimens. All shear test specimens after tempering or heat treatment shall meet the shear strength and metallurgical requirements of MIL-W-6858.

4.2 Tensile pullout specimens. All tensile pullout specimens after tempering in the welding machine shall have a minimum tensile strength not less than 25 percent of the shear strength requirements of MIL-W-6858.

4.3 Quality standards. The quality standards of the spot welds shall be in accordance with MIL-W-6858.

5. INSPECTION OF PRODUCTION PARTS

5.1 Unless otherwise specified, inspection of production parts shall be in accordance with MIL-W-6858.

5.2 Production check tests. Test specimens for checking the weld quality in production shall be made at the following specified intervals:

(a) At the beginning of each shift and at intervals not greater than 1 hour during production.

(b) Upon replacement of the electrodes.

(c) Each time a change is made in the welding schedule.

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5.3 Machine tempering. When spot welds are tempered by a heating cycle in the machine, the production check test shall consist of making and testing 3 shear specimens and 1 tensile pullout specimen.

5.4 Furnace tempering. When the spot welds are tempered or heat treated by a heating cycle in a furnace, the production check test shall consist of fracturing a weld in the as-welded condition and measuring the diameter of the fusion area. The weld diameter to the nearest 0.01 inch shall equal (within plus or minus 0.02 inch) the average weld diameter of the certification test specimens. In addition, three shear specimens welded and tempered with a batch of production items shall be made and tested at intervals of not more than 8 hours.

5.5 Macroscopic tests. Macroscopic specimens of two consecutive spot welds shall be sectioned through their centers for routine check purposes at intervals not exceeding 8 hours of production welding on one machine. Sections of spot welds are not mandatory each time the settings are changed, provided the settings are within the allowable variations from the settings used in the certification tests.

6. NOTES

6.1 Intended use. This specification covers additional requirements for spot welding hardenable steels that are not covered by MIL-W-6858.

6.2 This specification contains the requirements of MIL-W- 4994(USAF) dated 3 May 1955 which has been cancelled.

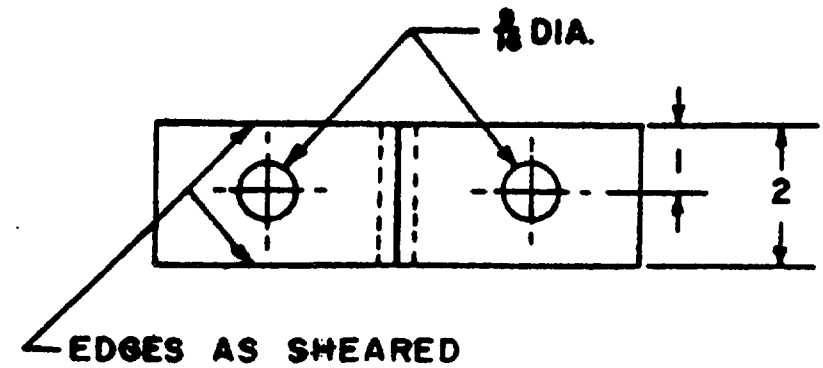
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Army - MR

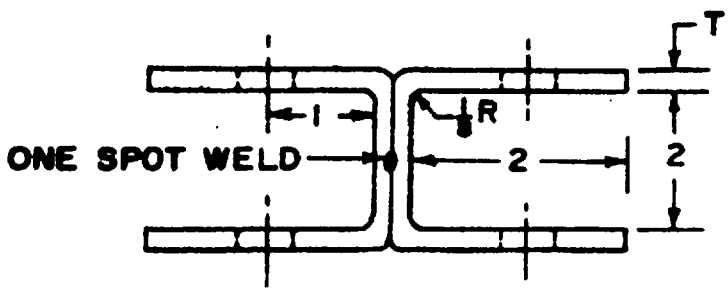
Preparing activity:
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Review activities:
Army - MI, WC, AT

Project No. THJM-A016

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64

FIGURE 1. TENSILE PULLOUT SPECIMEN

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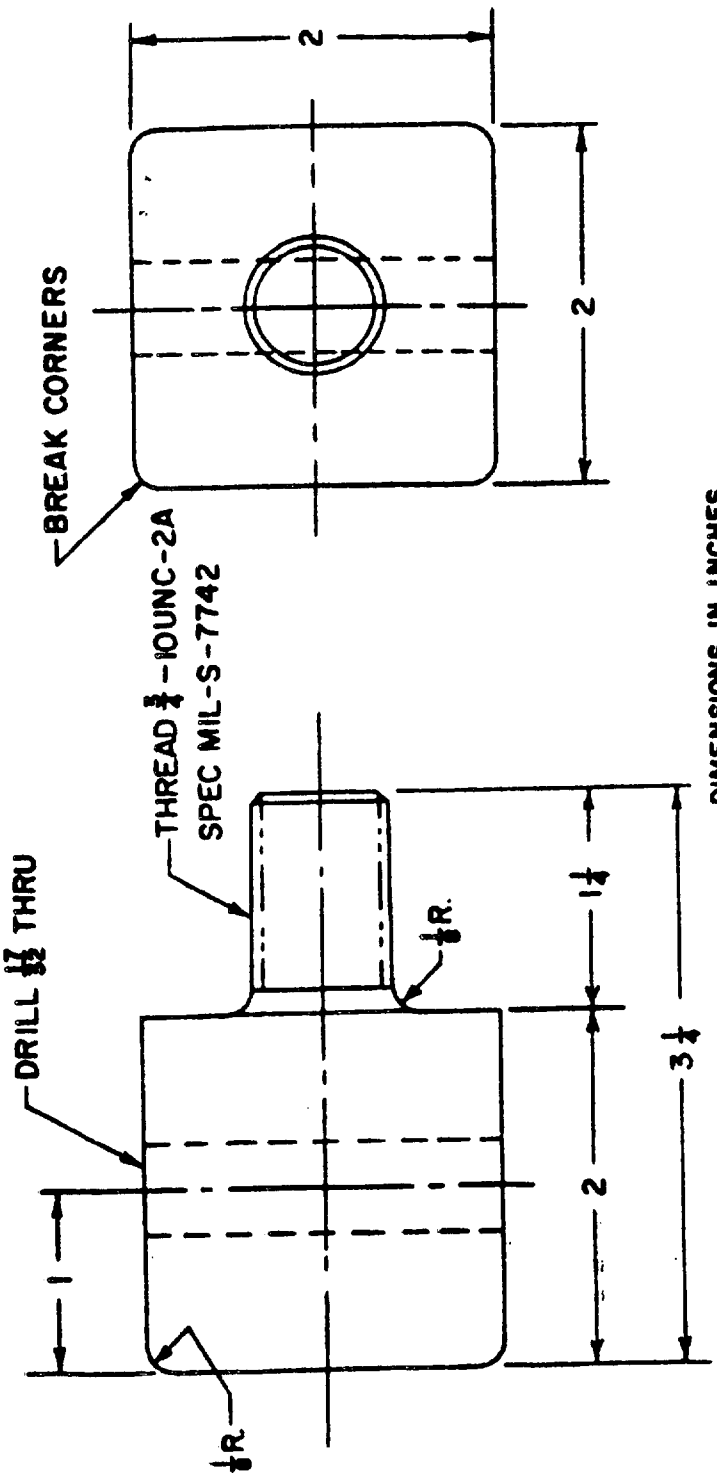


FIGURE 2. TENSILE PULLOUT JIG

☆U.S. GOVERNMENT PRINTING OFFICE: 1974—603-109/971

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.*)

SUBMITTED BY (*Printed or typed name and activity*) | DATE

DD, FORM, APR 62 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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